# Shard Heist

The game rules, for both rooms. Built from `docs/rules/*.md`.

# Game rules

> These documents describe what the game does, as implemented in the code. When a rule changes, update the document, the code and the tests together, then run `npm run docs` to rebuild the printable version.

The game has two rooms that are played one after the other. Until the group connects them, each room is its own page.

| Room | Document | Code with the rules | Tests |
|---|---|---|---|
| 1 · Plot Twist: Cell Block D (escape room) | room-1-escape-room.md | `room-1-escape/js/rules.js` | `tests/room-1/rules.test.js` |
| 2 · Shard Heist (police chase) | room-2-shard-heist.md | `room-2-chase/js/core.js` (rules), `room-2-chase/js/game.js` (pickups, police AI) | `tests/room-2/rules.test.js`, `tests/room-2/story.test.js`, `tests/room-2/input.test.js` |
| Both rooms: every setting and option | settings.md | `room-2-chase/js/core.js`, `room-1-escape/js/rules.js` (checks of stored values) | `tests/room-2/rules.test.js`, `tests/room-1/rules.test.js`, `tests/browser/browser.test.js` |

## The whole game in one paragraph

In April 2012 the Starfall necklace, lent by the Ashford family to Lisbon's naval museum for the centenary of the Titanic, vanished from its case. You were the detective on the case, until the trail ended at your own door. Now you are inmate 4471, convicted of stealing it. **Room 1:** you have ten minutes in a prison washroom to find the code for a hatch in the floor. If you get out (Ending A), a note shows that the guard wanted you out, and a car is waiting at the end of the tunnel. **Room 2:** the police chase begins. You drive a getaway vehicle through Lisbon's Baixa, on the highway and across the countryside, collecting star shards while up to four police cars and a helicopter hunt you.

## One night, one look

- **Time:** both rooms are set on the same night: room 1 around 02:15, room 2 from about 02:30.

- **Design:** both rooms share one design system: the same background, glass panels, fonts, buttons and cards. Each room keeps its own accent colour: **orange** in the prison, **cyan** on the run. Hale's radio messages in room 2 are orange because he comes from room 1.
- A test (`tests/project/page.test.js`) checks that the shared design tokens and the story links between the rooms stay in place.

## How the two rooms will connect (planned, not built)

- Ending A of room 1 shows a button that opens room 2.
- Ending B of room 1 stays in room 1 (try again).
- Open question for the group: does room 2 then start in free ride, or straight in the story mode *The Necklace Job*?

## Words used in these documents

| Word | Meaning |
|---|---|
| **Accepted action** | The player did something allowed. The state changes and is saved. |
| **Refused action** | The player tried something not allowed. Nothing changes, and the player is told why. |
| **State** | Everything the game remembers at one moment: where you are, items, time left, and so on. |
| **Scene** | A place you can stand in room 1 (the door, the sinks, ...). |
| **Hotspot** | Something you can use in a scene, shown as a pulsing marker (or, on a touch screen, a "Walk to" button). |
| **Tick** | One step of the clock, many times per second. |

# Room 1 · Plot Twist: Cell Block D

> This document describes the rules in `room-1-escape/js/rules.js`. Values marked **(open)** are not decided yet.

## Goal

Open the hatch in the floor of the toilet stall and climb out **before the clock reaches zero**.

## Story

The briefing is a short series of cards; **Next** and **Back** move between them.

1. **The front page.** *The Lisbon Times* of Monday 16 April 2012: the **Starfall necklace**, lent by the Ashford family to the Museu de Marinha for an exhibition marking 100 years since the Titanic sank, vanished from its glass case on the night of the centenary. No alarm sounded, and a detective was put on the case.
2. **Who you are.** Three weeks ago you were that detective. The trail ended at your own door: the necklace's velvet box was hidden in your car. Tonight you are **inmate 4471**, convicted of the theft: someone framed you. Your oldest friend on the force kept digging and found the proof that clears your name, but the people who framed you came after him. Before he disappeared from Lisbon, he hid the evidence across the city, in places only you would know to look.
3. **The door.** Officer Hale walks you to washroom 2. In a short scene in the room you turn round to the open door, see him in the corridor light, and watch him swing the door shut and lock it. Through the door he says: "Your ten minutes have begun. Not a second more." He waits outside.
4. **The rumour.** Inmates whisper about a hatch in the floor of this washroom, locked with a four-digit code. Three spots in the room hide a car key, the first two digits and the last two digits of the code, each behind a short Titanic mini game.
5. **How to play:** the controls.

The clock only starts when the player presses **Start the clock** after the briefing.

## Time

| Rule | Value |
|---|---|
| Time limit | **10:00** (open: 10 or 15 minutes; `TIME_LIMIT`) |
| Warnings from Hale | at 5:00, 2:00, 1:00 and 0:30, each only once |
| Wrong code | costs **30 seconds** (`WRONG_CODE_PENALTY`), never below 0:00 |
| Jammed door | when the clock hits 0:00 the first time, you get **one extra minute** (`WEDGE_GRACE`), once only |

| | |
|---|---|
| The clock stops | in the briefing, in the pause menu (which also opens when the browser tab is hidden or the window loses focus), and after an ending |
| The clock keeps running | while reading a clue, while playing a mini game and while turning the lock |

## Scenes and how to move

| Scene | What is there | You can walk to |
|---|---|---|
| **The Door** (start) | Steel door with a barred window, Hale outside; scratches in the wall beside it; mop in a bucket | Sinks, Stall |
| **The Sinks** | Two steel sinks, a polished metal mirror, an empty soap dish, a hot tap | Door, Stall |
| **The Stall** | Steel toilet, scratches on the wall, a crooked tile, the hatch in the floor | Door, Sinks, Hatch |
| **The Hatch** | The trapdoor with a four-wheel combination lock (you kneel down) | Stall |
| **The Tunnel** | Only reached through Ending A | – |

You walk freely through the washroom in the first person. Where you stand decides your scene: the stall (inside it or right in front of its doorway), the sinks (the west end of the room) or the door (everywhere else). You have to be well inside a new part of the room before it counts, so walking along a border never switches back and forth. The description of a scene is shown the first time you walk in. The Hatch is the only scene you do not walk into: in the stall you click the hatch (or its marker) to kneel down at the lock, and any step stands you up again. Walls, the sink counter, the bin, the bucket, the stall partitions, its open door and the toilet block your way.

On a laptop or desktop you only walk with the arrow keys; there are no buttons for it. A touch screen has no arrow keys, so there (and only there) "Walk to" buttons glide you to a fixed spot in another scene. Walking anywhere else is refused: "You cannot get there from here."

## Items (you can carry two)

| Item | Where | What it does |
|---|---|---|
| **Mop** | Bucket by the door | Optional. Used at the door ("Jam the door with the mop") it is placed under the door handle and leaves your inventory. This buys the extra minute. |
| **Car key** | In a hollow behind the crooked tile, once you have won the tile's mini game | Optional. A car key on a worn leather fob that someone hid there for you, a hint at the car waiting after Ending A. It opens nothing in this room. |

## Clues and the code

There are exactly **three clues**, one behind each of three spots. Clicking a spot opens a **mini game of at most 30 seconds**. Winning it gives the clue; losing it (or running out of the 30 seconds) gives

nothing, and the same game can be tried again at once. Giving up (Esc or **Give up**), pausing or an ending stops the game without a result.

| Spot | Where | Condition | Clue when won |
|---|---|---|---|
| **The crooked tile** | The Stall | – | The tile comes off and the **car key** lies behind it (take it with "Car key behind the tile") |
| **Writing on the mirror** | The Sinks | Only while the **hot tap** runs and steams the mirror up. Without steam you only see your face, and no game opens. | The **first two digits**: "UNSINKABLE? 19 _ _" appears in the steam |
| **Scratches by the door** | The Door | – | The **last two digits**: "_ _ 1 2" is gouged into the wall |

- The code is **1912**, the year the Titanic sank. It is the **same in every game**.
- A won clue is kept in your clue notes and can be read again at its spot; its mini game cannot be played again.
- The **graffiti** on the stall wall ("She was unsinkable. She sailed only once. The year opens the floor.") is a hint, not one of the three clues.

### The six mini games

At the start of every game, **three different games out of six** are dealt at random to the three spots (tile, mirror, scratches), so every round plays differently. The rules check that the three are different; the room deals them (`MiniGames.pickGames` in `room-1-escape/js/minigames.js`).

| Game | How to win in 30 seconds |
|---|---|
| **Battleship** | Two 6 × 6 grids, three ships each (3, 2 and 2 cells). As in the real game, ships never touch, not even at a corner; once a ship is sunk, the water around it is marked and cannot be fired at. Fire at the enemy waters; a hit lets you fire again. Sink all enemy ships before the bot sinks yours. The bot fires at random or, half the time, next to one of its hits. |
| **Spot the difference** | Two pictures of the Titanic at night; the right one has 5 changes, chosen from 9. Click all five, in either picture. A wrong click costs 2 seconds. |
| **Tic-tac-toe** | Your anchors against the bot's icebergs, you start. Three in a row wins, a draw starts a new round. The bot always takes a winning cell, blocks three times out of four, likes the centre, otherwise plays at random. |
| **Deck run** | Run along the boat deck to the bow; jump (Space, ↑ or a click) over deck chairs, crates, barrels and coils of rope. Three knocks and you are out. |
| **Deck climb** | Bounce up through the Titanic in five levels: the engine room (boilers, furnaces, pipes), the crew quarters (bunks and lockers), second class (oak-panelled corridors and the library), first class (the Grand Staircase with its clock and chandelier) and the boat deck with the funnels and lifeboats. Steer with ← → or the mouse. Lifebuoys bounce you higher. Falling back down loses. |
| **Seagull** | Fly the seagull over the sea, the Titanic in the background; flap with Space, ↑ or a click through ten gaps between storm clouds and icebergs. |

## The lock

- Four wheels, each showing 0–9. Up after 9 is 0, down after 0 is 9.
- You can only turn the wheels and try the code while kneeling at the hatch.
- **Try the code:**

- Right code: the hatch opens, Ending A. - Wrong code: "Clunk. The lock holds." Hale hears the noise, and you lose 30 seconds. The number of wrong codes is counted.

## Endings

| Ending | When | What happens |
|---|---|---|
| **A · Out below** | The right code before time runs out | You climb down the ladder. A note is taped to it: "Took you long enough. A car is waiting at the end of the tunnel. — H." The guard wanted you out. That is the plot twist that leads to room 2. |
| **B · Back to the cell** | The clock reaches 0:00 (after the extra minute, if the door was jammed) | The door bangs open and Hale and two officers take you back to cell 4471. |

**The earlier decision:** jamming the door early gives one more minute at the very end. That can turn Ending B into Ending A.

After an ending, every action is refused except **Play again / Try again**, which starts a new game from the briefing.

## Refused actions (nothing changes, the player is told why)

| The player tries to... | Message (short) |
|---|---|
| do anything before the briefing is finished | "Read the briefing first." |
| walk where there is no way | "You cannot get there from here." |
| take an item that is not here, or twice | "There is no ... here." / "You already have the ..." |
| jam the door without the mop, or a second time | "You need something long and solid..." / "The door is already jammed." |
| take the car key while the tile is still on | "The tile is still in place." |
| play the mirror's mini game while it is dry | No game opens: "Just your own tired face in the scratched metal..." |
| play a mini game that is already won, or from another place | "You already have this clue..." / "You cannot reach that from here." |
| read a clue before its mini game is won | "You have not won this clue yet. Try the mini game first." |
| use the tap away from the sinks, the lock away from the hatch | "There is no tap here." / "Kneel down at the hatch to reach the lock." |

| | |
|---|---|
| turn a wheel that does not exist | "The lock only has four wheels." |
| do anything after an ending | "The story is over. Start a new game to play again." |

## Saving

- The game is saved in the browser after **every accepted action** and every few seconds while the clock runs.
- After a reload, the player can **Continue** (with the time that was left) or start a **New game**, which asks for confirmation first.
- A saved game is checked field by field when loaded. Anything that is not a real save is ignored, and a new game starts.

## Controls

| Input | Action |
|---|---|
| Arrow keys (or `W A S D`) | Walk: up and down forwards and backwards, left and right sideways |
| Drag the mouse or a finger | Look around, all the way round |
| Click a pulsing marker or the object itself | Use it (the tile, the mirror and the scratches open their mini game) |
| Space, arrow keys, mouse | Play a mini game (see each game); while a game is open you do not walk |
| "Walk to" buttons (touch screens only) | Glide to another scene |
| "Clues" button (bottom left) | Read your clue notes |
| `P` / `Esc` | Pause (Esc also closes a card) |
| `M` | Sound on or off |
| Volume slider | On the pause card, 0 to 100 %. It is kept for the next visit |
| `Enter` | Confirm the main button on a card |

# Room 2 · Shard Heist

> This document describes the rules in `room-2-chase/js/core.js`, and the pickups and police in `room-2-chase/js/game.js`.

## Where room 2 starts

Room 2 begins where room 1 ends. You (inmate 4471, the framed detective) have climbed out of the tunnel under Cell Block D. It is about 02:30 at night. In a lock-up garage, Officer Hale's getaway vehicle is waiting. Every police car in town is now hunting the escaped inmate.

**Officer Hale is a secret ally.** He believes you were framed, but he cannot prove it. In the story mode he helps over the radio; his messages are shown in **orange**, the colour of room 1. Police dispatch messages are **red**.

## Two ways to play

- **Free ride:** pick a map and a getaway vehicle, then escape.
- **Story mode,* The Necklace Job:* three acts on the three maps, in order, with a story and three endings.

## Vehicles

| Vehicle | Top speed | Acceleration | Handling | Damage taken | Ramming |
|---|---|---|---|---|---|
| **Car** (muscle coupe) | 100 % | 100 % | 100 % | 100 % | 1.0 |
| **Motorcycle** (superbike) | 112 % | 130 % | 140 % | 100 % | 0.4 |
| **Truck** (tractor unit) | 78 % | 70 % | 72 % | **55 %** | 2.4 |

Percentages are relative to the map's top speed and to the car. The menu shows the same values as bars (1–5) and as km/h and 0–80 km/h times, computed from these numbers.

## Maps

| Map | Goal | Time | Car top speed | Police top speed | Shards | Repair kits |
|---|---|---|---|---|---|---|
| **Lisbon · Baixa** | Collect **all 6 pieces of evidence**, pick up the **witness**, then drive through the golden gate that appears at the south-west corner of the Baixa (Rua do Crucifixo at the river avenue), on the way to the 25 de Abril bridge | 3:00 | 122 km/h | 119 km/h | 6 pieces of evidence | 3 |

| | | | | | | |
|---|---|---|---|---|---|---|
| **Highway** | Reach the finish **4 km** ahead. Shards are optional: **+4 s** each | 1:50 | 200 km/h | 187 km/h | 12 | 2 |
| **Countryside** | Reach the village at the end of the road. Shards: **+4 s** each | 2:30 | 158 km/h | 148 km/h | 10 | 3 |

- A shard or a piece of evidence is picked up within 3.8 m; a repair kit within 3.6 m, and only when you have damage.
- Lisbon has traffic lights, 26 other cars, two white city buses, four motorcycles and three yellow trams. The trams only run on their tracks, a stone track bed with two rails set into the road: a clockwise loop through Rua da Conceição, Rua da Prata, the north side of the Praça do Comércio and Rua Áurea. As on real Lisbon streets, cars and the player can drive over the tracks. Each wheel that rolls over a rail gives a jolt: the picture and the vehicle shake, the rail clacks, and phones that can vibrate give a short buzz. The faster you go, the harder the jolt; the front wheels hit first and the rear ones a moment later. Driving along the stone track bed rumbles lightly. The trams themselves run on steel railway wheels with a flange on the inside, on a two-axle truck with springs and brake blocks. The buses are 12 m long, so at a crossing they go straight on or turn left in the wide arc, never right into the tight corner (unless that is the only way left at the edge of the map). Everything stops at red lights and keeps its distance to the back of the vehicle ahead, so nobody runs into the long tail of a bus or a tram. On the highway traffic drives both ways: cars, SUVs, lorries, three coaches and six motorcycles; lorries and coaches keep to the middle and slow lanes. On country roads there is oncoming traffic, with one coach and three motorcycles among it, and driving off the road cuts your top speed to 55 %.
- **Other vehicles at night:** every vehicle throws its headlights on the road ahead, so a car waiting at a dark crossing can be seen from far away. Its bright brake lights are on while it slows down or stands. You can crash into buses and motorcycles like into any other vehicle.
- Each map has an edge you cannot cross. When you near it the screen turns red and a "turn back" arrow appears. In Lisbon the outer streets are part of the map, so there this only happens when you head for the edge.

## The evidence in Lisbon

In Lisbon you do not collect star shards. Your oldest friend on the force found out who really broke into the naval museum and hid the proof across the Baixa (room 1 tells this in its briefing). There are six pieces, one of each, in random places every run. Each floats above the street with a glow, and the panel at the top left shows one icon per piece that lights up when it is in the car.

| Evidence | What it proves |
|---|---|
| **Photos** | A figure at the museum window at 01:40 on the night of the theft, while you were at home |
| **Fingerprints** | Fingerprints and DNA traces from the display case, and they are not yours |
| **Torch** | The thief's torch, dropped in the museum garden, with paint from the window frame on it |
| **Lockpick** | The museum's lock was opened with these picks, not forced |
| **Clothes** | The dark clothes from the photos, with glass splinters from the display case in the seams |
| **Balaclava** | The balaclava from the photos, with hairs inside that are not yours |

**Before you drive** a map of the Baixa shows where the six pieces lie, with their names, the repair kits, your start and the gate. The clock waits until you press **Start driving**. **While you drive** an arrow on the screen (at the bottom in the middle; on an upright phone under the pursuit panel) points to the nearest piece still out there, with its name and distance; once all six are found it turns gold and points to the witness, and once she is in the car, to the gate. (The story's sealed letter is on neither.)

**The witness.** With all six pieces in the car, one thing is still missing: a person. **Marta Silva**, a night-shift nurse, saw the real thief climb out of the museum window, and she will say so in court. She waits on a pavement somewhere in the Baixa, 70 to 200 m from where you are, with a golden light over her head. She turns to face your car and waves: now and then while you are far away, and without stopping once you are closer than 45 m. The radar shows her in gold (an arrow at the rim while she is out of range), the signal bars point to her, and a seventh icon appears in the panel. Stop next to her, closer than 7 m and slower than about 11 km/h, and she runs to the passenger door on the right. Once the vehicle stands still, she opens the door, steps in (up the steps into the truck's cab), sits down and pulls the door shut; on the motorcycle she climbs onto the pillion seat. Getting in takes about one second, and the vehicle waits for her. From then on she rides along beside you, visible through the window, over the bridge as well. Drive off before she reaches the vehicle (more than 12 m away) and she goes back to wait. With her on board, the golden gate opens at the south-west corner of the Baixa, where Rua do Crucifixo meets Avenida Infante Dom Henrique by the river. It faces the 25 de Abril bridge, and driving through it takes you onto the bridge.

**The story card.** As soon as Marta has shut the door, the run pauses and a card in the middle of the screen tells what she saw and where to go now. Nothing moves until you close it (the button, `Enter`, or a tap); the clock and the police wait as well.

**The yellow arrows.** From then on yellow arrows lie on the road, one every 5 m in your lane, and show the way through the streets to the gate, round every corner. They are worked out again as you drive, so after a wrong turn they simply show a new way. A pulse runs along them towards the gate. The police keep chasing you while you wait for her.

**The finale.** Drive through the gate and the picture fades: you are on the **25 de Abril bridge** with Marta beside you, driving south over the Tejo at about 86 km/h while the first light of the morning comes up in the east. There is nothing more to do: no police, no helicopter, heroic music (brass, strings and timpani, about 14 seconds, on the music volume). Then the result card. The drive over the bridge does not count towards your time. In the story, it ends act 1.

The highway and the countryside keep their **bonus shards** (+4 s each).

## The city map: Lisbon's Baixa

The Baixa is the centre of Lisbon, rebuilt as a street grid after the 1755 earthquake. The map is a simplified version of it: the blocks are square, and the grid has 7 streets each way. The real Baixa has a few more, so some are left out. The real Baixa is also flat; the hills on this map are a game decision.

| Direction | Streets, in their real order |
|---|---|
| West to east | Rua do Crucifixo, Rua Áurea, Rua dos Sapateiros, **Rua Augusta**, Rua dos Correeiros, Rua da Prata, Rua dos Fanqueiros |
| North to south | Rua de Santa Justa, Rua da Vitória, Rua de São Nicolau, Rua da Conceição, Rua do Comércio, Praça do Comércio (north side), Avenida Infante Dom Henrique |

- In free ride, the panel at the top left shows the street or square you are on. In story mode it shows the act instead.
- You start on Rua Augusta, facing south towards the Arco da Rua Augusta.
- **Squares:** the Rossio and the Praça da Figueira in the north, the Praça do Comércio by the river in the south. The statue on the Praça do Comércio stands in the middle of the road, so traffic passes on both sides.
- **Landmarks you can see but not reach:** the river Tejo, the 25 de Abril bridge (you only cross it in the finale), the castle hill, and across the river the town of Almada: a quay, an ochre cliff and on top of it about 1,250 houses in the pastel colours of Lisbon: small two-storey houses with shutters, town houses with shops and iron balconies, and a few apartment blocks, with hipped tile roofs and windows that are lit at night. The houses stand in rows along streets, with umbrella pines and cypresses between them and all over the cliff. Above it stands Cristo Rei, the floodlit statue of Christ with outstretched arms on its tall four-pillar pedestal, facing Lisbon. The river moves: long swells and fine ripples, it mirrors the sky towards the horizon, and the lights of the far bank tremble on it as long streaks. The Santa Justa lift stands on the map, and you can crash into it.
- **Parks:** there are three, which the real Baixa does not have. The story needs them for the sealed letter.
- **Houses:** closed blocks of Pombaline houses, wall to wall, as the Baixa was rebuilt after 1755: a ground floor plus 3 to 5 floors (OpenStreetMap gives most buildings 5 levels), arched stone ground floors with old shop fronts, the tallest windows and the balconies on the first floor, limewash colours, and tile roofs with dormers. Tiled fronts are the exception, as they are in the real Baixa.
- **Real buildings in their place:** the National Theatre and Rossio station north of the Rossio, the yellow arcades and corner towers of the Praça do Comércio, and the Sé, seen at the end of Rua da Conceição. Their position is simplified, and they are modelled freely, not measured.
- It is always about 02:30 at night. The clock on the arch shows it.

### How the map was made

The city is a hand-built model of the real Baixa, not map data loaded by the game.

1. **Reference.** We used OpenStreetMap (© OpenStreetMap contributors, ODbL) as the reference. From it we took the names of the streets and squares, the order in which they follow each other from west to east and from north to south, where the squares and the river front lie, the route of the tram loop, and the usual height of the buildings (most have five levels).
2. **Simplified layout.** The real Baixa is a rectangular grid, rebuilt after the 1755 earthquake, so it fits a game well. We drew it as a regular grid of 7 × 7 streets: square blocks of 48 m and streets 14 m wide. The real blocks are longer than they are wide and vary in size, and there are a few more streets, so the order is real and the measurements are not.
3. **Written into the code.** The street names and the grid are fixed in `room-2-chase/js/core.js` (the `BAIXA` table, with `streetAt` naming the street under the car). The game ships no map file and calls no map service or API while it runs. It needs no key and keeps working if a map service is down.
4. **Landmarks modelled by hand.** The Arco da Rua Augusta, the Praça do Comércio, the Rossio with the National Theatre and the station, the Santa Justa lift, the Sé, the castle hill, the 25 de

Abril bridge, Cristo Rei and Almada were modelled from how they look and placed where they roughly stand, not from measured plans. Some are changed on purpose: the bridge has a main span of 300 m instead of 1,013 m so it fits the view, the hills and the three parks were added for the game, and the finish gate stands at the south-west corner, towards the bridge.

## Driving

| Rule | Value |
|---|---|
| Top speed | map top speed × vehicle speed, **minus up to 40 %** at 100 % damage |
| Throttle | never pushes you above the top speed |
| Brake | slows you down, then reverses, up to 40 km/h backwards |
| No pedal | the car rolls to a stop |
| Handbrake (`Space`) | slows you quickly and turns sharper |
| Headlights | always on: a wide, soft beam that lights the road ahead and fades into the dark |
| Brake lights | on while you brake, in either direction, and with the handbrake while moving |
| Reversing lights | white, on while you roll backwards |
| Skid marks and tyre squeal | with the handbrake above about 15 km/h, when braking above about 30 km/h, and in very fast corners (above about 80 km/h, steering hard). They only mark the road, and they are cleared when a new run starts |

**Hills.** All three maps have hills, up to about 20 % steep. The Arco, the river front and the two ends of the highway are level. Uphill your top speed falls by up to 16 % and gravity pulls you back, half as strongly as it pushes you downhill. Downhill your top speed rises by up to 22 % and you coast on. The truck feels slopes most, the motorcycle least. Brakes and rolling friction hold a stopped vehicle in place.

**Smooth picture.** If the game runs below 48 frames per second for 2 seconds, it lowers the resolution one step (85, 70, then 60 %). If that does not help, the step is taken back. After 20 seconds at 57 or more it goes back up, at most twice. The choice of Ultra or Balanced stays yours.

## Damage

| Rule | Value |
|---|---|
| Crash | impacts under 4 m/s (about 14 km/h) are free; above that, damage = (impact − 4) × 1.0 × the vehicle's damage factor. A wall at 122 km/h costs the car 30 %, the truck 17 %. One crash counts once: for 0.35 s after contact, only a harder hit costs more |
| 25 % damage | the paint shows dents (deepest at the nose and the tail), long scratches and dirt, and loses its shine as the damage grows; thin grey smoke rises from the engine |
| 50 % damage | warning: "Engine damaged: top speed reduced." Loose parts and a smashed light; the engine misfires; the smoke turns dark |

| | |
|---|---|
| 75 % damage | hood up (car), screen gone (motorcycle), hanging bumper and cracked windscreen (truck); a thick black column of smoke; sparks at speed |
| 88 % damage | the engine is on fire |
| **100 % damage** | **vehicle wrecked: game over** |
| Repair kit | removes 40 % damage, and the damage you see goes with it |

## Police

| Rule | Value |
|---|---|
| First police car | after 4 seconds; in the countryside once you are about 170 m along the road |
| **Wanted stars** | 1 to 5, shown under the clock. A new star every fifth of the time limit or of the way to the end (evidence in the city, distance on the open roads), whichever is further along. Each wrecked police car counts as 4 % more. Stars never fall. |
| New police car | until the number for your stars is reached, see the table below |
| Ramming | every hard hit lowers a police car's health by impact × your ramming power × 0.09; a police car is wrecked when its health is gone and comes back after the time in the table below |
| **Police wreck** | stays where it stopped: lights out, dented and dull, sagging to one side and smoking for about 14 s. Traffic stops behind it, and you and the police can crash into it. It is cleared when you are more than 260 m away, or after 30 s once you are more than 70 m away. When that car is sent back into the chase, it comes back clean |
| Far behind | a police car more than 140 m away drives 20 % faster; one more than 230 m away in the city, or 320 m on the open roads, is sent back near you with full health |
| Hills | police slow down uphill and speed up downhill as you do; other traffic keeps its speed |
| **Bust meter** | fills while a police car is **within 9 m** and you are **slower than 4.5 m/s** (about 16 km/h): full in 2.2 s. Otherwise it drains (empty from full in 3 s). |
| **Busted** | bust meter at 100 %: game over |

| Stars | Police cars | Police speed | New car every | Wrecked car returns after | Health |
|---|---|---|---|---|---|
| ★ | 1 | 100 % | 3 s | 9 s | 3 |
| ★★ | 2 | 100 % | 3 s | 9 s | 3 |
| ★★★ | 3 | 103 % | 3 s | 8 s | 3 |
| ★★★★ | 3 | 106 % | 2.5 s | 6 s | 3.5 |
| ★★★★★ | 4 | 110 % | 2 s | 4.5 s | 4 |

**Helicopter.** After 60 seconds on the run (or sooner with the fifth star), a police helicopter joins the chase for the rest of the run and hunts you with its searchlight. Its searchlight follows you at 94 % of the police speed. While you are in the light (within 9 m of its centre), the police drive 6 % faster, and in the city they see you from 170 m instead of 110 m. It cannot bust you or damage you. An undamaged car or motorcycle at full speed on a clear, level road stays ahead of the light; the truck, and a badly damaged car, cannot.

## Radio

Seven stations, or off: Reggae, Techno, Hardstyle, Schlager, Sing-Along Pop, Classical and Heavy Metal. Choose in the menu, or at any time with `R` or the radio button. The music plays while you drive, and is silent in the menu and in pause.

All of it is composed and synthesised by the game, with no music files. The tunes are generated by the game. The classical station plays Bach and Beethoven, whose works are in the public domain.

## Ending a run

| Result | When |
|---|---|
| **Clean getaway** | Lisbon: reach the gate under the arch, with the six pieces of evidence and the witness; the drive over the bridge follows. Highway: pass the 4 km mark. Countryside: reach the village |
| **Busted** | the bust meter is full |
| **Wrecked** | damage reaches 100 % |
| **Time's up** | the clock reaches 0:00 |

## Stars and best times

- **Drive time.** In the city this is the time used. On the highway and in the countryside it is the time used plus the bonus seconds the shards gave you, so collecting shards never makes a run look slower.
- **Stars:**

| Map | ★★★ | ★★ | ★ |
|---|---|---|---|
| City | more than 40 % of the time left | more than 20 % left | otherwise |
| Highway | under 88 s | under 105 s | otherwise |
| Countryside | under 115 s | under 135 s | otherwise |

- These limits are for the car. For the other vehicles the drive time counts multiplied by their speed factor (motorcycle 1.12, truck 0.78), so each can earn three stars.
- More than 50 % damage costs one star, but never the last one.
- The best drive time per map is saved in the browser. Saved values under 10 s or over 15 minutes are ignored, because no run can take that long or that little.

## Drivers and the leaderboard

- **Driver.** In the main menu (section 03) you create a driver by typing a name, or pick a saved one. A name keeps letters, digits, spaces and `. _ ' -`, up to 16 characters; capital letters do not matter, so *alex* and *Alex* are the same driver.
- **Leaderboards.** Every finished free ride (Lisbon, Highway, Countryside) goes on two boards of its map under the active driver: the **all-time** board and the board of the **current calendar month**. Each driver has one entry per board, their best time; a slower run never replaces it. The fastest time is first, and on a tie the time that was there first stays ahead. When a new month begins, the monthly board starts empty; the all-time board stays.
- **Leader of the month.** The fastest driver of the current month on a map is named above the two boards.
- **Where they are shown.** In the main menu (for the map that is chosen) and on the end screen after the finish. Each board shows the top five, and the driver's own place below them if they are further down. The end screen says whether the run is a new personal best and which place it has on both boards. It is set in large type: the drive time is the one big number, with the best time on this device under it, the stars beside the title, and time left (or shards), damage and wrecked police cars next to it. Without a driver, the end screen asks for a name first; the time is saved as soon as the name is entered.
- Runs that end busted, wrecked or out of time, and the story mode, do not go on the board.
- The drivers and the boards are kept in this browser only. There is no password and nothing is sent anywhere, so the board shows the people who play on the same device.

## Story mode: The Necklace Job

| Act | Map | Goal | Special rule |
|---|---|---|---|
| **1 · The Evidence** | Lisbon | Recover the 6 pieces of evidence, pick up the witness, then the gate and the bridge | An optional **sealed letter** is hidden in a park. It is **not shown on the radar**. |
| **2 · The Courier** | Highway | Catch the gold-marked courier lorry (get within 9 m) to take the **clasp**, then the finish | The **toll at the finish only opens with the clasp**: without it you stop there |
| **3 · Homecoming** | Countryside | Drive back to the Ashford mansion | Officer Hale waits on the right verge 70 m before the finish and calls out when you are close. The confrontation decides the ending |

| Ending | When |
|---|---|
| **A · The truth** | You found the letter: it proves Lady Ashford staged the theft. Hale gives you your badge back. |
| **B · Back in 4471** | No letter: the evidence alone, brought by an escaped convict, is not believed. Lady Ashford hands you to the police, and you go back to cell 4471. |

| | |
|---|---|
| **C · Caught** | Busted, wrecked or out of time in any act: back to Cell Block D. You can retry that act. |

**Retrying an act:** a letter found during the failed attempt does **not** count. A letter found in an earlier, finished act does.

Radio messages (Hale in orange, police dispatch in red) appear at fixed times in each act, each only once.

## Menu actions (accepted and refused)

| Button | Allowed when |
|---|---|
| Start the getaway | in the main menu |
| Story mode | in the main menu, or after the final ending |
| Start act 1 | on the story's opening card |
| Drive again / Restart | after a run, or in the pause menu |
| Resume | in the pause menu |
| Main menu | anywhere except while driving |
| Next act | after a finished act, but not after the last one |
| Retry act | after Ending C, or in the pause menu during a story |

Any other button or action is refused, and no card button works while you are driving.

## Saved settings

The last map and vehicle, the best times, the drivers and leaderboards, the radio station, the graphics, the steering and both volumes are saved in the browser. Every saved value is checked when loaded; invalid values fall back to Lisbon, Car, radio off and the defaults. All settings are explained in Settings and options.

## Controls

| Input | Action |
|---|---|
| `W A S D` or arrow keys | Drive |
| `Space` | Handbrake |
| `P` / `Esc` | Pause |
| `M` | The large map of the Baixa (Lisbon only); `M` again closes it. A tap on the radar does the same |
| `N` | Sound on or off |
| Volume sliders | In the menu and on the pause card: `Game` (engine, effects, sirens) and `Music` (the radio), 0 to 100 %. They are kept for the next visit |

| | |
|---|---|
| `R` | Next radio station (`Shift` + `R` for the one before) |
| `Enter` / `Space` | Press the button in focus. With no button in focus, `Enter` presses the card's main button. While driving, `Space` is the handbrake |
| `Tab` | Move between the buttons of a card |
| Touch pads | Steer on the left; brake and gas on the right. Both together are the handbrake |
| Tilt (phones) | Hold the phone sideways and lower one side, like a steering wheel: from 3° it starts to steer, at 24° it is at full lock. The steering buttons go away while tilt works. Upright, or if the sensor is refused or missing, the buttons stay. Choose `Tilt` or `Buttons` in the menu or on the pause card. The sensor is only read in the browser and nothing is stored or sent |

# Settings and options

> Every setting the player can change, in both rooms: where it is, what each choice does, and what the game remembers. All of it is stored only in this browser ( `localStorage` ). Nothing is sent anywhere. The only "account" is a driver name for the leaderboard, with no password, kept on this device.

## Room 2 · Shard Heist: the main menu

| Setting | Choices | What it does |
|---|---|---|
| **Map** | Lisbon (Baixa), Highway, Country Roads | Where the run takes place. Each map has its own goal, time limit and top speed (see *Maps* in the room 2 rules) |
| **Vehicle** | Car, Motorcycle, Truck | Speed, acceleration, handling, how much damage a crash does and how the police can push you. The card under the choice shows the numbers |
| **Mode** | Start the getaway (free ride), or the story card *The Necklace Job* | Free ride is one run on the chosen map. The story plays all three maps in order, with Hale on the radio and three endings |
| **Radio** | Off, Reggae, Techno, Hardstyle, Schlager, Sing-Along Pop, Classical, Heavy Metal | The station that plays while you drive. `R` changes it at any time, `Shift` + `R` goes back one |
| **Graphics** | Ultra, Balanced | **Ultra:** sharper picture (up to twice the screen resolution), smoothed edges, glowing lights and the finest shadows (4096 × 4096). **Balanced:** a lighter picture without the glow and with smaller shadows (2048 × 2048). Computers start in Ultra, phones in Balanced |
| **Steering** (phones only) | Tilt, Buttons | **Tilt:** hold the phone sideways and lower one side like a steering wheel. **Buttons:** steer with the arrow pads. Tilt is the default on a phone with a motion sensor |
| **Driver** | The saved drivers, and a field for a new name | Creates a driver (just a name, up to 16 letters, digits, spaces and `. _ ' -` ) or picks a saved one. Each finished free ride goes on the leaderboard of its map under the active driver. Names are matched without case, so *alex* and *Alex* are the same driver. Up to 12 drivers are kept; adding a 13th drops the oldest from the list (its times stay on the boards) |
| **Volume** | `Game` 0–100 %, `Music` 0–100 % | `Game` is the engine, tyres, sirens, the helicopter and all effects. `Music` is the radio. The music plays for a moment while its slider moves, so it can be heard; letting go of the game slider plays a short tone |

## Room 2: during a run

| Setting | How | What it does |
|---|---|---|
| **Pause** | `P`, `Esc` or the pause button | Stops everything: the clock, the police, the traffic and the music. The pause card shows the shards, time, damage and bust meter |
| **Pause card** | Resume, Restart (or Retry act in the story), Main menu | It also has the steering choice (phones) and both volume sliders |
| **Map** (Lisbon) | `M`, or a tap on the radar | Pauses the run and shows the large map of the Baixa: where you are, the evidence, repair kits and witness that are still out there, the gate, and the yellow way to it once it is open. `M` or `Enter` closes it |
| **Sound** | `N` or the speaker button | Turns all sound off or on, without changing the volume sliders |
| **Radio** | `R` or the radio button | Next station; the name is shown under the clock |

## Room 2: what the game sets by itself

| Setting | Rule |
|---|---|
| **Resolution** | If the game runs below 48 frames per second for 2 seconds, the resolution drops one step: 85 %, then 70 %, then 60 %. A step that does not make the game faster by at least 5 % is taken back and the setting stays. After 20 seconds at 57 frames per second or more it goes one step back up, at most twice. Choosing Ultra or Balanced starts again at full resolution |
| **Shadows** | The area that gets shadows (120 m wide) follows the player in whole shadow pixels, so shadow edges do not crawl over the walls |
| **Phones** | Phones start in Balanced and use shadows of half the size |
| **Less motion** | If the system asks for reduced motion, the camera and the vehicle do not shake after crashes or on the tram rails, the radar sweep stops and the menu turns more slowly |
| **Vibration** | On a phone whose browser can vibrate (most Android phones; iPhones cannot), each tram rail under a wheel gives a short buzz of 12–30 ms, at most one every 0.09 s |
| **Hidden tab** | When the tab is hidden or the window loses focus, the run pauses and the sound stops |

## Room 1 · Plot Twist: Cell Block D

| Setting | How | What it does |
|---|---|---|
| **Continue or New game** | Shown after a reload, if there is a saved game | Continue goes on where you were, with the time that was left. New game asks first, then deletes the save |
| **Pause** | `P`, `Esc` or the pause button | Stops the clock. The pause card has Resume, Restart and the volume slider |

| | | |
|---|---|---|
| **Volume** | Slider on the pause card, 0–100 % | All sounds of the washroom: the light's hum, dripping water, the tap and the effects. A short tone plays when it is let go |
| **Sound** | `M` or the speaker button | All sound off or on |
| **Walk** | Arrow keys or `W A S D` | Walks through the washroom in the first person |
| **Look around** | Drag the mouse or a finger | Turns your head, all the way round |

## What is remembered, and what happens to bad values

Every stored value comes from outside the program, so each one is checked when it is loaded. A value that is missing or wrong is ignored and the default is used; the game never breaks because of it.

| Stored | Room | Default if missing or invalid |
|---|---|---|
| Map and vehicle | 2 | Lisbon and the car |
| Best time per map | 2 | No best time (only 10 s to 900 s counts) |
| Drivers and leaderboards (all time, and the current month) | 2 | Empty boards; names are cleaned again, impossible times and unknown maps are dropped, only each driver's best run per map is kept, and a monthly board with an invalid month is dropped |
| Active driver | 2 | No driver: the end screen asks for a name |
| Radio station | 2 | Off |
| Graphics | 2 | Ultra on a computer, Balanced on a phone |
| Steering | 2 | Tilt |
| Game and music volume | 2 | 100 % each; a value outside 0–100 % is cut to the range |
| Saved game | 1 | No saved game; any save that no real game could produce is refused |
| Volume | 1 | 100 % |

To reset everything, clear this site's data in the browser settings.